جامعة محمد بن زايد للذكاء الاصطناعي
MOHAMED BIN ZAYED UNIVERSITY OF ARTIFICIAL INTELLIGENCE

presight
A G42 company

SemEval 2023

61 ACL 2023

BERTastic at SemEval-2023 Task 3:

Fine-Tuning Pretrained Multilingual Transformers – Does Order Matter?

Tarek Mahmoud – tarek.mahmoud@mbzuai.ac.ae
Preslav Nakov – preslav.nakov@mbzuai.ac.ae

Contributions

1. The first to observe a human-like primacy-recency effect (aka serial position effect) (Murdock, 1962) in deep learning models in general and in transformers in particular.
2. The first to propose and to evaluate sequential fine-tuning of transformers language by language.
3. Our proposed method outperforms the naïve approach by achieving F1 score improvements of **2.57%**
4. Even when we supplement the naïve approach with recency fine-tuning, we still achieve an improvement of **1.34%** with a **3.63%** convergence speed-up.

Naïve Approach

EU Profits From Trading …
Comment Macron et sa …
Es gibt viele gute Gründe, …
Si può legittimamente …
Eksploatowanie wątków …
Американская пропаганда …
…
Kreml organizuje dla …

A sample minibatch

Sequential Fine-Tuning Approach (Ours)

Fine-Tune on **English**
↓
Fine-Tune on **French**
↓
Fine-Tune on **German**
↓
Fine-Tune on **Italian**
↓
Fine-Tune on **Polish**
↓
Fine-Tune on **Russian**

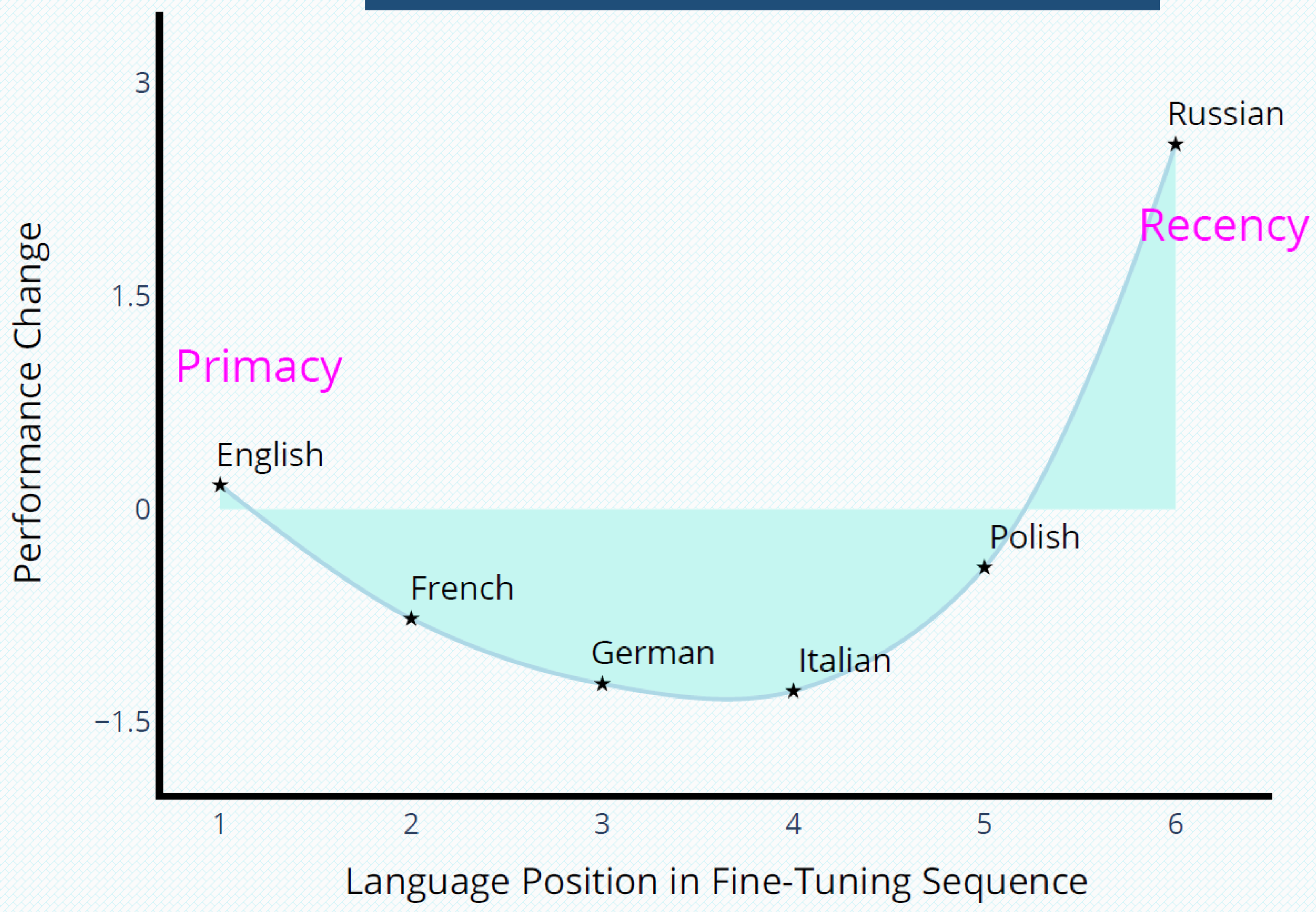
Intuitively, we expect highest performance on languages we fine-tune on last (recency effect)

However, contrary to intuition, we also witness an accuracy improvement on languages in the first position

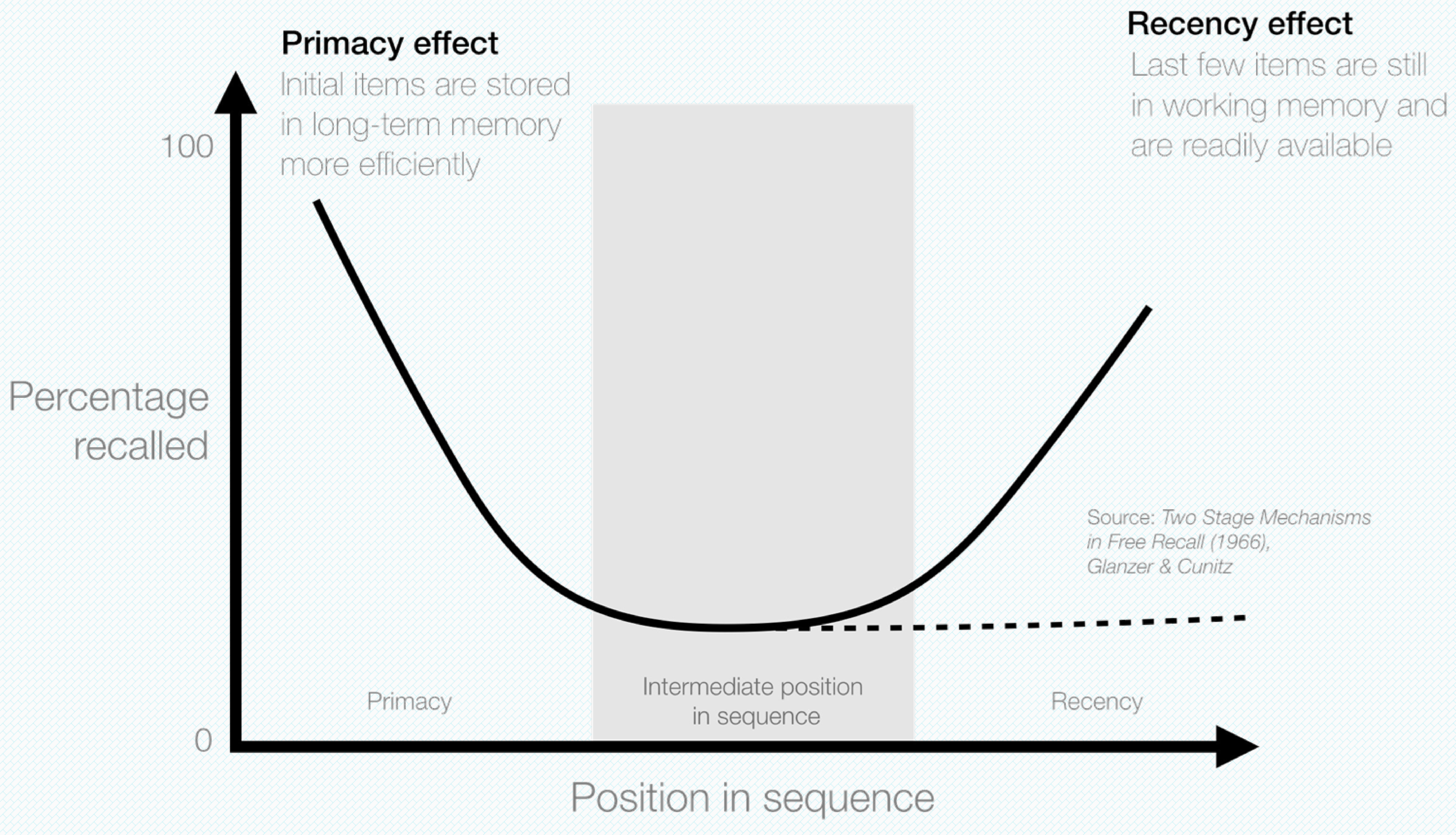
The news-framing dataset has 6 languages
→ There are 6! = 720 possible fine-tuning sequences

Primacy-Recency Effect

Deep Learning (Ours)



Human Brain



Credits: https://ui-patterns.com/patterns/Serial-positioning-effect

Results

Method	English	French	German	Italian	Polish	Russian	Average
XLM-R-BL	47.11	36.39	49.70	43.44	52.00	32.03	43.45
XLM-R-S	63.98	55.45	61.99	**56.33**	63.07	46.01	57.81
XLM-R-S-FT*	65.91	57.4	**63.25**	52.53	65.11	50.00	59.03
XLM-R-O*	**70.21**	**58.51**	61.27	53.75	**65.96**	**52.53**	60.37
$\Delta_{XLM-R-S}$	6.23	3.06	−0.72	−2.58	2.89	6.52	2.57
$\Delta_{XLM-R-S-FT*}$	4.30	1.11	−1.98	1.22	0.85	2.53	1.34

XLM-R-O*	Number of Examples	XLM-R-S-FT*	Number of Examples	Speed Up
XLM-R-O1	15522	XLM-S-FT-RU	15405	−0.76%
XLM-R-O2	15458	XLM-S-FT-EN	18963	18.48%
XLM-R-O3	14336	XLM-S-FT-FR	15450	7.21%
XLM-R-O4	15385	XLM-S-FT-IT	15884	3.14%
XLM-R-O5	15684	XLM-S-FT-PO	15556	−0.82%
XLM-R-O6	16071	XLM-S-FT-GE	15240	−5.45%
			Average Speed Up	3.63%